



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.AT.11

Apply Two-Variable Relationships to Solve Problems

Lesson 9-4 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can write an equation to represent the relationship between two variable quantities and use it to solve a real problem.

Language Objective: I can justify a recommendation using the words equation, solution, and at least.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

solution

substitute

at least

predict

justify



Tap any word to see its meaning and a picture.

1

A value of the variable that makes an equation true is a

2

To replace a variable with a number so you can compute is to

3

That amount or more — the smallest amount that still works — is

4

To use what you already know to say what a future amount will be is to

5

To explain WHY your answer makes sense using the mathematics is to

 it.

Write About the Math

From which bakery would you recommend Yelina purchase cupcakes? Explain?

¿De qué panadería le recomendarías a Yelina comprar los panecitos? Explica?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **solution**
solución

☐ **substitute**
sustituir

☐ **at least**
al menos

☐ **predict**
predecir

☐ **justify**
justificar

Model: Since h is being multiplied by 2.5 in the equation, dividing by 2.5 is the operation that undoes the multiplication and isolates h . — A strong answer names division as the inverse of multiplication and checks the solution by substituting it back in. A weak answer just says 'because that's what you do' without the inverse-operation reasoning.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Write an ____ for each bakery, substitute Yelina's \$ ____ budget into each one, ____ the two results, and recommend the bakery that gives more ____ for the money.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** solution
2. **Notes 2:** substitute
3. **Notes 3:** at least
4. **Notes 4:** predict
5. **Notes 5:** justify
6. **Watch for this mistake:** *Adding the rate instead of multiplying — writing $2.5 + 8 = 10.5$ cars for an 8-hour shift instead of $2.5 \times 8 = 20$. The rate applies to EVERY hour, so the total is always rate $\times$ amount.*
7. **Try It 1:** A. No — cars come in whole numbers, so round up to the next whole car — *You cannot wash seven-tenths of a car, so the model's decimal answer must be interpreted: round up so the goal is still met.*
8. **Try It 2:** A. Either can work: \$12 needs at least 42 cars washed, while \$14.30 needs only about 35 — so the choice depends on how many customers they expect — *Both plans reach \$500 — $42 \times 12 = \$504$ and $35 \times 14.30 = \$500.50$ — so the real question is whether they can attract 42 cars at the lower price, or whether the higher price might turn customers away.*